Model	Knowledge Cutoff
GPT-4o-20240806	Sept 30, 2023
GPT-4o-mini-20240718	Sept 30, 2023
GPT-4.1-2025-04-14	May 31, 2024
GPT-4.1-2025-04-14	May 31, 2024
o3-2025-04-16	May 31, 2024
o3-pro-2025-06-10	May 31, 2024
Claude-3-5-sonnet-20241022	April 30, 2024
Claude-3-5-sonnet-20240620	April 30, 2024
Qwen3-32B	June 2024
Qwen3-235B-A22B	June 2024
DeepSeek-v3-0324	July 1, 2024
DeepSeek-R1-0528	July 1, 2024

Table 2: Knowledge cutoff dates for various language models

My dataset has questions about events that would happen in the world, ranging from technological advancements, politics, and movie awards. Example questions can be found in Appendix A. I use a news pipeline, called AskNews, which crawls almost all news articles on the web. They use Llama 3.1-72B to summarize the article in a few sentences. AskNews returns 30 articles relevant to the question, going as far back as 60 days from the open date of the question, and I feed each question with its associated news articles to the LLM. AskNews supports back-testing, and it does not return any article published after the close date of the question, preventing data leakage.

Additionally, I also have a separate set of 130 binary questions of events that occur between October 2024 to December 2024. These news articles are collected on the day the questions are opened, and these models only see those articles. These events all occur after the models' knowledge cut-offs. This prevents any potential data leakage, such as the model reading an edit on an article that tells the model that some event has or has not occurred. The models do not perform significantly differently on this 200-question set over the 334-question set, where news articles were collected after question resolution. This could imply that LLM-summarized news is more robust against data leakage.

To minimize noise and inherent variations in the language model, I ask the model to make five predictions on the same question with the same prompt at the default temperature. I then report the naive mean and median. OpenAI reasoning models had the reasoning set to "medium," the default. Each prediction is a separate API call, and the models do not have access to what they have predicted in the past. I use the average of the five predictions to compute the Brier score, a common measurement of prediction accuracy. Averaging the Brier score reduced the noise from outlier predictions, and I also report the median predictions. The formula for the Brier score is the mean squared error, defined as:

$$BrierScore = \frac{1}{N}\sum_{i=1}^{N}(f_i - o_i)^2$$

where f_i is the prediction given by the model, and o_i is the resolution of the question. A resolution of $o_i = 1$ means the question resolved as "yes," and an $o_i = 0$ means that the question resolved as "no."

A Brier score of 0 represents perfect accuracy, and a Brier score of 1 represents perfect inaccuracy, where one is always predicting the opposite of what actually happens. If one predicts 50% for every question, the Brier score should be 0.25. Getting a score larger than 0.25 implies that one is doing worse than random. Older LLM models, such as GPT-3.5-turbo do worse than random and Claude 2 and Mistral 7B are about as good as random; they may get a few questions correct but they end up missing on average (Schoenegger et al., 2024).

Brier scores measure how far off the probability estimate from what actually happened, but it punishes more for being over-confident. It is a proper scoring method, because a forecaster is rewarded for predicting their true estimate instead of over- or under-estimating. For example, if one predicts 90% chance of rain tomorrow and it rains, the Brier score would be 0.01 as $(0.9 - 1)^2 = 0.1$. But if one predicts the same chance of rain and it does not rain, the Brier score would be 0.81 as $(0.9 - 0)^2 = 0.81$. This person's average Brier score in a scenario if it rained and did not rain on different days would be 0.455. They would have been much better off at predicting a 60% chance of

rain, the Brier score would be 0.16 if it rained and 0.36 if it did not rain. The average Brier score would be 0.26 (note that a 50% prediction on everything would yield a Brier score of 0.25). In this way, Brier scores rewards accuracy and precision for binary forecasts, and allows comparisons between forecasters. The early versions of GPT-4 models tend to hover around 0.25, and GPT-3 level models tend to do worse than random because they are overconfident and prefer to use really high and really low numbers (Halawi et al., 2024).

I also break down the questions in my dataset into seven categories: arts & recreation, economics & finance, environment & energy (weather), healthcare & biology, politics & governance, science, and sports. Gemini 2.5 Flash classified all the questions. The prompt is in figure 7.

Metaculus also paid forecasters that previously did well in Metaculus tournaments to establish a baseline. They predicted on 47% of the questions that the bots predicted every day, making a data set of 157 questions. On the held-out dataset, the expert forecasters predicted on 41 questions, about 31% of the dataset.

5 Results

5.1 Direct Prediction

The Brier score for o3 is 0.1352, better than the human crowd forecasting score of 0.149 found in (Halawi et al., 2024) but worse than the 0.121 score found in (Karger et al., 2025). Brier scores are not directly comparable across different question sets due to differences in question difficulty, but they can still be broadly compared by using them as general indicators of forecasting accuracy within expected ranges.

Model	Median Ensemble		Mean Ensemble	
	Score	Std Error	Score	Std Error
o3	0.1352	0.0097	0.1362	0.0097
o3-pro	0.1386	0.0099	0.1389	0.0099
GPT-4.1	0.1542	0.0139	0.1509	0.0132
o4-mini	0.1589	0.0123	0.1589	0.0120
Deepseek-v3	0.1798	0.0115	0.1761	0.0109
Claude-3.6-Sonnet	0.1810	0.0126	0.1785	0.0120
GPT-4o	0.1883	0.0118	0.1851	0.0113
Qwen3-235B-A22B	0.1923	0.0122	0.1931	0.0122
Deepseek-r1	0.1950	0.0126	0.1914	0.0119
Claude-3.5-Sonnet	0.1947	0.0125	0.1910	0.0119
Qwen3-32B	0.2066	0.0129	0.2041	0.0123
GPT-4o-mini	0.2743	0.0143	0.2676	0.0133

Table 3: Median and mean Brier scores for direct prediction

GPT-4.1 comes next at 0.1542 and o4-mini follows close behind at 0.1589. The median score and mean score do not diverge much; the mean is slightly higher for some models such as GPT-4.1 but lower for other ones like o3. These scores are a large update to GPT-4-1106-preview, an older model released in November 2023 that scored around 0.20 (Halawi et al., 2024). Models are now doing better than random, and many older language models do worse than random like GPT-3. Open-source models still lag behind frontier AI models as well. Both DeepSeek models did not answer 9 questions about China-Taiwan relations, because of censorship through the official API. Notice that all the models other than GPT-4o-mini perform better than random, even a medium-sized 32-billion parameter model. Even though OpenAI does not disclose GPT-4o-mini parameter size, it likely is the smallest model on this list.

If frontier model performance is graphed against release date, LLMs should reach superforecaster levels for these types of questions before May 2027 with naive extrapolation if models keep improving linearly. Open-source models are not shown in this graph because they lag closed-source by a few months to a year.

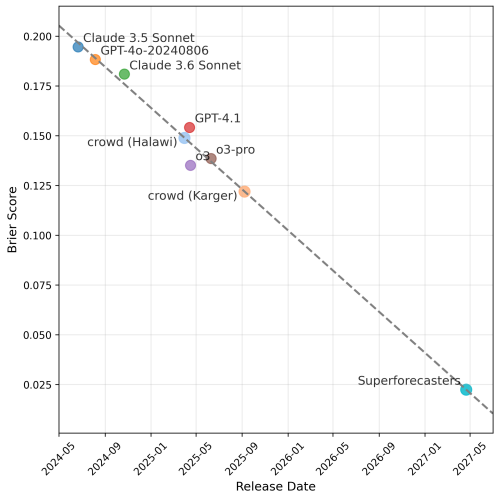

Figure 1: Brier Scores and Date of Release

When analyzing the score by category, all the models have better scores for politics and governance than economics and finance. Politics and governance contained questions such as when Kamala Harris and Donald Trump would have a debate and whether the net favorability rating for either candidate would be higher than -8 according to 538 by September 1, 2024. Economics and finance contained questions about the CPI, unemployment rate, and whether certain companies would file for bankruptcy before a certain date. Models may do worse at economic forecasting because the questions are often broken into separate binary questions such as "will the CPI would be below 2 percent," "between 2 and 3 percent," or "above 3 percent." LLMs tend to do worse with numbers, and this could be related as well. Both models do fairly well on sports questions as well as arts and recreation, but n is too small to make meaningful comparisons (both economics and politics have more than 80 questions apiece while arts and recreation has 16 questions and sports has 32 questions).

Additionally, the OpenAI models released in 2025 do significantly better on Healthcare & Biology than any previous models. An example question is in Table 12. There are only 31 questions in this category, which is smaller than "Economics & Finance" or "Politics," so results should be taken with a grain of salt.

Model	Arts & Rec	Economics	Environ.	Healthcare	Politics	Sci & Tech	Sports
Claude-3.5-Sonnet	0.1385	0.2052	0.2286	0.1747	0.1815	0.2382	0.1836
Claude-3.6-Sonnet	0.1328	0.2039	0.2018	0.1656	0.1611	0.2173	0.1693
DeepSeek-v3	0.1414	0.1933	0.2343	0.1428	0.1799	0.1869	0.1631
DeepSeek-r1	0.1525	0.1896	0.2689	0.1913	0.1782	0.2499	0.1737
GPT-4.1	0.1477	0.1740	0.2435	0.0819	0.1283	0.2080	0.1463
GPT-4o	0.1223	0.1947	0.2947	0.1500	0.1688	0.2448	0.1777
GPT-4o-mini	0.1417	0.2593	0.2280	0.2886	0.2956	0.3273	0.2389
o3-pro	0.1279	0.2818	0.2850	0.1391	0.1655	0.2011	0.1970
o3	0.1740	0.1353	0.1893	0.0921	0.1199	0.1486	0.1649
o4-mini	0.1804	0.1668	0.2245	0.1005	0.1380	0.2024	0.1640
Qwen3-235B	0.1454	0.1943	0.2259	0.1481	0.1874	0.2316	0.1989
Qwen3-32B	0.1501	0.2170	0.2416	0.1777	0.1968	0.2322	0.2141

Table 4: Median Brier scores for models by category with direct prediction

This trend holds in both the newer and older models. Newer models do better at predicting economic events, but they still do not do as well on those questions compared to politics. They do very well on questions relating to healthcare and biology, which contained questions like will the CDC report more than 80% of the tested influenza sequences as influenza A during the 2024-25 season through the week ending Dec 21, 2024.